



2026

Chemistry

Paper 3 — Electrochemistry · Rates of Reaction · Energy Changes

Time allowed: 45 minutes

Total marks: 50

Index Number: _____

Instructions to Candidates

1. Write your Index Number in the space provided above.
2. Answer **all** questions in **all three sections**.
3. Section A: Choose **one** answer for each question. Tick the box next to your chosen answer.
4. Section B & C: Write your answers in the spaces provided.
5. Use a blue or black pen. Diagrams may be drawn in pencil.
6. All working must be shown for calculations.
7. The number of marks is given in brackets [] at the end of each question.

Section	Marks Available	Marks Awarded
A	15	
B	20	
C	15	
Total	50	

SECTION A — Multiple Choice [15 marks] Answer all 15 questions. Tick ONE box per question.

- 1 Electrolysis is the decomposition of an ionic compound (in molten or aqueous form) by [1]
- ☐ A heat energy
- ☐ B electrical energy (passing an electric current)
- ☐ C light energy
- ☐ D chemical energy from a reaction
- 2 During electrolysis, reduction occurs at the [1]
- ☐ A anode (positive electrode)
- ☐ B cathode (negative electrode)
- ☐ C electrolyte solution only
- ☐ D both electrodes simultaneously
- 3 During the electrolysis of molten lead(II) bromide (PbBr_2), what is produced at the cathode? [1]
- ☐ A Lead (Pb) metal
- ☐ B Bromine (Br_2) gas
- ☐ C Hydrogen gas
- ☐ D Oxygen gas
- 4 A catalyst increases the rate of a chemical reaction by [1]
- ☐ A increasing the temperature of the reaction
- ☐ B increasing the concentration of reactants
- ☐ C lowering the activation energy of the reaction
- ☐ D providing extra energy to the reactant particles
- 5 Which of the following factors does **NOT** affect the rate of a chemical reaction? [1]
- ☐ A Temperature
- ☐ B Concentration of reactants
- ☐ C Surface area of a solid reactant
- ☐ D Colour of the reactants
- 6 As the temperature of a reaction mixture increases, the rate of reaction [1]

- ☐ A increases because particles have more energy and collide more frequently and with greater energy
- ☐ B decreases because particles slow down
- ☐ C remains unchanged
- ☐ D first increases then decreases

7 Powdered marble (CaCO_3) reacts faster with dilute acid than lumps of marble of the same mass because [1]

- ☐ A powder has greater mass
- ☐ B powder has greater surface area exposed to the acid, so more frequent collisions occur
- ☐ C powder contains more calcium carbonate
- ☐ D powder has a lower activation energy

8 An exothermic reaction is one that [1]

- ☐ A releases energy (heat) to the surroundings
- ☐ B absorbs energy from the surroundings
- ☐ C has no overall energy change
- ☐ D only occurs at high temperatures

9 On an energy profile diagram, the activation energy is represented by [1]

- ☐ A the energy of the products
- ☐ B the difference in energy between the reactants and the peak (transition state)
- ☐ C the difference in energy between the reactants and the products
- ☐ D the energy released when products form

10 During the electrolysis of aqueous copper(II) sulfate solution using copper electrodes, which of the following describes what happens at the **anode**? [1]

- ☐ A Copper is deposited on the anode
- ☐ B Hydrogen gas is produced at the anode
- ☐ C Copper from the anode dissolves into the solution
- ☐ D Oxygen gas is produced at the anode

11 Which of the following reactions is endothermic? [1]

- ☐ A Combustion of natural gas
- ☐ B Thermal decomposition of calcium carbonate
- ☐ C Neutralisation of an acid with a base
- ☐ D Rusting of iron

- 12 The energy released when one mole of a substance burns completely in oxygen is called the [1]
- ☐ A heat of neutralisation
 - ☐ B heat of formation
 - ☐ C activation energy
 - ☐ D heat of combustion
- 13 The overall enthalpy change (ΔH) for an exothermic reaction is [1]
- ☐ A positive
 - ☐ B zero
 - ☐ C negative
 - ☐ D equal to the activation energy
- 14 Electrolysis is used industrially to purify copper. This is because electrolysis with copper electrodes [1]
- ☐ A converts copper into copper oxide at the cathode
 - ☐ B causes pure copper to be deposited at the cathode while impure copper dissolves at the anode
 - ☐ C removes impurities by burning them off at high current
 - ☐ D causes copper to be deposited at the anode
- 15 When the concentration of a reactant in solution is increased (at constant temperature), the rate of reaction increases because [1]
- ☐ A the activation energy is lowered
 - ☐ B there are more reactant particles per unit volume, so collisions are more frequent
 - ☐ C particles move faster
 - ☐ D a catalyst is produced in the reaction

SECTION B — Short Structured Questions [20 marks] Answer all questions.

B1 Electrolysis of molten lead(II) bromide

The diagram below shows the apparatus used for the electrolysis of molten lead(II) bromide (PbBr_2).

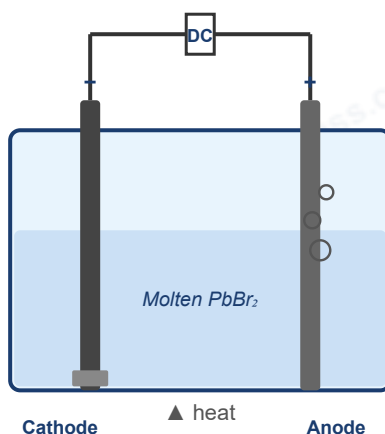


Fig. 1 — Electrolysis of molten lead(II) bromide

- (a) Why must lead(II) bromide be **molten** (not solid) for electrolysis to work?

[1]

- (b) Name the product formed at the cathode and at the anode. State whether each change is oxidation or reduction.

[2]

Cathode product:

Anode product:

- (c) Write the ionic half-equation for the reaction at the cathode.

[2]

B2 Rates of Reaction

A student investigates the effect of concentration on the rate of reaction between sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$) and hydrochloric acid. The mixture goes cloudy as sulfur is produced. She measures the time taken for a cross drawn under the flask to become invisible.

Experiment	Conc. of $\text{Na}_2\text{S}_2\text{O}_3$ (mol/dm^3)	Time / s
1	0.10	120
2	0.20	60
3	0.40	30
4	0.80	15

- (a) Describe the relationship between concentration and rate of reaction shown by this data. [1]

- (b) Explain, using collision theory, why increasing concentration increases the rate of reaction. [2]

- (c) State **one** variable that must be kept constant in this experiment to make it a fair test. [1]

- (d) The word equation for this reaction is:
sodium thiosulfate + hydrochloric acid $\rightarrow$ sodium chloride + sulfur dioxide + sulfur + water
Name the gas produced and state how you could test for it. [1]

B3 Energy Changes in Reactions

Fig. 2 shows the energy profile diagram for a chemical reaction.

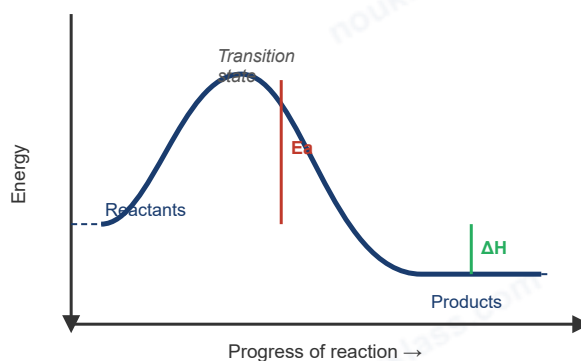


Fig. 2 — Energy profile diagram

- (a) State whether the reaction shown in Fig. 2 is exothermic or endothermic. Give a reason using Fig. 2.

[2]

- (b) On the diagram above, mark with a letter **C** where a catalyst would place the peak of the curve. Explain what effect this has on the rate of reaction.

[2]

- (c) State whether the value of ΔH for an exothermic reaction is positive or negative. Explain why.

[1]

B4 Electroplating

A manufacturer wants to electroplate a steel spoon with silver. The diagram shows the electrolysis cell used.

- (a) State what material the cathode should be and what material the anode should be for silver electroplating.

[2]

Cathode:

Anode:

(b) Name a suitable electrolyte for silver electroplating.

[1]

(c) Describe what happens at the cathode during electroplating. Write the half-equation for this reaction.

[2]

SECTION C — Long Structured Questions [15 marks] Answer all questions.

C1 Electrochemistry — Electrolysis of Aqueous Solutions

[8]

A student electrolyses aqueous copper(II) sulfate (CuSO_4) solution using (i) inert carbon (graphite) electrodes, then (ii) copper electrodes.

(a) Using inert (carbon/graphite) electrodes:

- (i) Name the product at the cathode and write the ionic half-equation for its formation. [2]

- (ii) Name the product at the anode and write the ionic half-equation for its formation. (Note: O_2 is produced in dilute solution.) [2]

(b) Using copper electrodes:

- (i) State what is observed at the cathode during this electrolysis. [1]

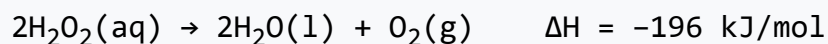
- (ii) State what happens to the **anode** during this electrolysis and explain why the concentration of Cu^{2+} ions in the solution remains approximately constant. [2]

- (c)** State **one** industrial use of the electrolysis of copper(II) sulfate with copper electrodes. [1]

C2 Rates of Reaction and Energy Changes

[7]

Hydrogen peroxide (H_2O_2) decomposes to form water and oxygen. The reaction is slow at room temperature but is speeded up by manganese(IV) oxide (MnO_2) as a catalyst.



- (a) State, with a reason, whether the decomposition of hydrogen peroxide is exothermic or endothermic. [1]

- (b) Explain how MnO_2 increases the rate of decomposition, using ideas about activation energy. [2]

- (c) After the reaction is complete, the student recovers the same mass of MnO_2 that was added. What does this tell you about MnO_2 in this reaction? [1]

- (d) A student measures the volume of oxygen produced every 30 seconds. Sketch a graph on the axes below to show how the volume of oxygen changes with time for (i) no catalyst, and (ii) with MnO_2 catalyst. Label each curve. [2]



- (e) State **one** other method (not a catalyst) that could be used to increase the rate of decomposition of hydrogen peroxide. Explain how it works using collision theory.

[1]

Mark Scheme — Paper 3

For teacher/self-assessment use only. Half-equations: must be balanced for charge and mass. Accept alternative correct half-equations. State symbols required only where asked. Credit correct scientific reasoning even if wording differs.

Section A — Multiple Choice

Q	Answer	Q	Answer	Q	Answer
1	B	6	A	11	B
2	B	7	B	12	D
3	A	8	A	13	C
4	C	9	B	14	B
5	D	10	C	15	B

Section B

Question	Mark-worthy points	Marks
B1(a)	When solid, ions are in fixed positions and cannot move / carry charge; when molten, ions are free to move and carry the current	1
B1(b)	Cathode: lead (Pb) metal — reduction [1]; Anode: bromine (Br ₂) gas — oxidation [1]	2
B1(c)	$\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$ [1 for correct species, 1 for balancing charge]	2
B2(a)	As concentration increases, the rate increases / time decreases; doubling concentration halves the time (doubles the rate)	1
B2(b)	More particles per unit volume [1]; so more frequent collisions between reactant particles; more collisions with energy $\geq$ activation energy [1]	2
B2(c)	Any one: temperature; volume of acid; volume of sodium thiosulfate solution; same viewing conditions	1
B2(d)	Sulfur dioxide (SO ₂); test: hold damp blue litmus paper / damp universal indicator paper at mouth of tube — turns red/acid; OR bubble into acidified potassium permanganate — decolourises	1
B3(a)	Exothermic [1]; products are at lower energy than reactants (on diagram) / energy is released [1]	2
B3(b)	C marked below original peak on diagram [1]; lower activation energy means more particles have energy $\geq E_a$ so more successful collisions per second — rate increases [1]	2
B3(c)	Negative [1]; energy of products is lower than reactants — energy is released to surroundings so the system loses	1

Question	Mark-worthy points	Marks
	energy [1] (accept: $\Delta H = H_{\text{products}} - H_{\text{reactants}}$, which is negative for exothermic)	
B4(a)	Cathode: steel spoon (object to be plated) [1]; Anode: silver (pure silver) [1]	2
B4(b)	Silver nitrate solution (AgNO_3) or silver cyanide solution	1
B4(c)	Silver ions (Ag^+) from the electrolyte gain electrons at the cathode and are deposited as silver metal on the spoon [1]; half-equation: $\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag}(\text{s})$ [1]	2

Section C

Question	Mark-worthy points	Marks
C1(a)(i)	Copper deposited at cathode [1]; $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$ [1]	2
C1(a)(ii)	Oxygen gas produced at anode [1]; $2\text{H}_2\text{O}(\text{l}) \rightarrow \text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^-$ [1]	2
C1(b)(i)	Copper metal is deposited (pink/copper-coloured layer builds up) on the cathode	1
C1(b)(ii)	Copper anode dissolves / decreases in mass [1]; $\text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$ — copper enters solution at same rate as it is deposited at cathode, so $[\text{Cu}^{2+}]$ stays constant [1]	2
C1(c)	Purification / refining of copper (pure copper obtained at cathode; impurities fall to bottom as anode sludge)	1
C2(a)	Exothermic; ΔH is negative (-196 kJ/mol) / energy is released	1
C2(b)	MnO_2 provides an alternative reaction pathway with lower activation energy [1]; more H_2O_2 molecules have energy $\geq$ the (lower) E_a at room temperature, so the rate increases [1]	2
C2(c)	MnO_2 is a catalyst — it is not consumed in the reaction; it is chemically unchanged at the end	1
C2(d)	Two curves both levelling off at the same final volume (same total O_2) [1]; catalyst curve rises more steeply and levels off sooner [1]	2
C2(e)	Increase temperature [1]; higher temperature gives particles more kinetic energy, so more collisions per second and greater proportion exceed the activation energy [1] (accept: increase concentration of H_2O_2 — more particles per unit volume, more frequent collisions)	1